Closed differentiation and signed adjoints for Schwartz densities with zeros

Project Stardust
A classical formalization companion

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Abstract

For every real Schwartz density K, differentiation on Schwartz functions is densely closable in $L^2(|K|\,dx)$, without a square-integrable logarithmic derivative. We give a direct distributional construction using bounded coefficients and a weight-squared embedding, keeping the original graph closure distinct from its closed weak extension. Explicit total adjoint probes and multiplication by the sign of K identify the exact domain of the signed adjoint. Orthogonal graph projection constructs the positive Hilbert energy resolvent, its exact minimization gap and the full nonnegative self-adjoint energy operator. Dyadic resolvents approximate the original graph on its domain, while native outputs escape at every excluded limit. An auxiliary Gaussian field has exactly this resolvent as its covariance, with a proved Green identity and original first-chaos source calculus. The logarithmic derivative is always absolutely integrable against $|K|\,dx$ but fails to be square-integrable at a simple zero. A normalized sign-changing polynomial-Gaussian example exhibits this failure while retaining all closed operators. An integrating factor also excludes polynomial first-order Stein weights for sign-changing kernels. The mechanisms are classical weighted operator and indefinite-inner-product theory; no new general closability or spectral principle is claimed. A separate Gaussian/Dirac mixture shows that full support does not replace the density hypothesis.

1 The original spaces and conclusions

Fix a real Schwartz function $K \in \mathcal{S}(\mathbb{R};\mathbb{R})$ and write $w = |K|$, $\mu(dx) = w(x)\,dx$. No positivity, nonvanishing, normalization or finite-zero hypothesis on K is imposed. The measure μ is finite. Throughout,

$$H = L^2(\mu;\mathbb{C}), \qquad \langle u, v\rangle = \operatorname{Re}\int \overline{u}v\,d\mu$$

is regarded as a *real* Hilbert space. All adjoints and self-adjointness assertions below refer to this real structure, matching the formal sources. Distribution tests are complex Schwartz functions with complex bilinear distribution evaluation. The zero-density case gives the zero Hilbert space and is included.

Let D_0 take the H class of a Schwartz function f to the class of f'. Its representative independence will be proved, rather than presumed. The principal result is the following.

Theorem 1.1. *The operator D_0 is densely defined and closable. Its graph closure D is densely defined and closed. Set $Ju = \sigma u$, where $\sigma = 1$ on $\{K \geq 0\}$ and $\sigma = -1$ on $\{K < 0\}$. The signed adjoint is the densely defined closed operator*

$$D^+ = JD^*J, \qquad \operatorname{dom} D^+ = \{v \in H : Jv \in \operatorname{dom} D^*\}.$$

For the nondegenerate signed real form $[u,v] = \operatorname{Re}\int \overline{u}vK\,dx$, it satisfies

$$[Du, v] = [u, D^+v] \quad (u \in \operatorname{dom} D, v \in \operatorname{dom} D^+),$$

and this pairing condition characterizes its whole domain and value. The Hilbert energy operator

$$A = D^*D, \qquad \operatorname{dom} A = \{u \in \operatorname{dom} D : Du \in \operatorname{dom} D^*\}$$

is densely defined, closed, nonnegative and self-adjoint. Its resolvent $R = (I + A)^{-1}$ is a bounded symmetric positive contraction, injective with dense range. For every $x \in H$ and $u \in \operatorname{dom} D$,

$$\begin{aligned}\|x - u\|^2 + \|Du\|^2 = \|x - Rx\|^2 + \|DRx\|^2 \\ + \|u - Rx\|^2 + \|Du - DRx\|^2.\end{aligned} \tag{1}$$

Thus Rx is the unique minimizer of the displayed energy.

The positive operator A uses the *Hilbert* adjoint D^*, not the signed adjoint D^+. Positivity means its Hilbert quadratic form is nonnegative; it is not an assertion of order preservation or a probabilistic transition kernel. Neither operator is identified with a time generator of a signed pseudoprocess.

2 A bounded-coefficient distributional rigging

For $u \in H$ and $\varphi \in \mathcal{S}(\mathbb{R};\mathbb{C})$ define

$$Iu(\varphi) = \int \varphi u\,d\mu, \qquad Eu = I(wu), \qquad Fu = I(bu),$$

where

$$b(x) = 2\sigma(x)K'(x), \qquad w(x)b(x) = 2K(x)K'(x). \tag{2}$$

Both w and b are bounded measurable functions, since K, K' are Schwartz. In particular these formulas define continuous real-linear maps from H into the complex tempered distributions. For example Cauchy–Schwarz gives

$$|Eu(\varphi)| \le \|u\|_H \|w\varphi\|_H, \qquad |Fu(\varphi)| \le \|u\|_H \|b\varphi\|_H,$$

and each second factor is at most a fixed constant times $\sup|\varphi|$. Thus every evaluation is continuous in u; the bounds also give the usual distribution topology continuity. No quotient by K appears.

It is important to retain the extra factor in E:

$$Eu(\varphi) = \int \varphi u\, w^2\,dx = \int \varphi u\, K^2\,dx. \tag{3}$$

This is not the distribution represented by u against Lebesgue measure, which need not exist locally across zeros. Its density is integrable, because $\int |u|w^2 \le \|u\|_H(\int w^3)^{1/2} < \infty$. The map E is injective. Indeed $Eu = 0$ implies $uw^2 = 0$ Lebesgue almost everywhere by separation of locally integrable densities by smooth compactly supported tests. One elementary justification of that separation is to convolve locally with a compactly supported smooth approximate identity: all convolutions are zero on smaller intervals, and converge to the density in local L^1. Hence the density is zero there, and a countable interval covering gives the assertion on $\mathbb{R}$. Since $w > 0$ μ-almost everywhere, it follows that $u = 0$ in H.

For a Schwartz input f, integration by parts gives the exact identity

$$E(f') = (Ef)' - Ff. \tag{4}$$

To check all signs, evaluate the right side on φ. It is $-\int \varphi' f K^2 - \int 2\varphi f K K'$, which equals $\int \varphi f' K^2$. Every term is absolutely integrable, and the boundary terms vanish because the factors are Schwartz. This uses the smooth weight K^2, even though $|K|$ need not be differentiable at a transverse zero.

3 The weak graph and the original derivative closure

Define

$$\mathcal{G} = \{(u, v) \in H \times H : Ev = (Eu)' - Fu\}. \tag{5}$$

It is a closed real linear subspace: for a convergent sequence (u_n, v_n) in it, continuity of each test evaluation passes the identity to the limit. If $(0, v) \in \mathcal{G}$, injectivity of E gives $v = 0$. Thus $\mathcal{G}$ is the graph of a closed partially defined operator D_{w}.

Equation (4) shows that every pair (f, f') of Schwartz classes lies in this graph. In particular if a Schwartz f is zero as an H class, then its derivative class is zero: D_0 is well-defined on the range of Schwartz functions in H. This range is dense. For clarity, finite Radon measure regularity first approximates indicators of measurable sets in L^2 by continuous compactly supported functions: choose compact and open inner/outer approximations with small measure difference and a continuous cutoff between them. Truncation and simple-function approximation reduce a general L^2 function to these indicators. Finally mollification approximates a continuous compactly supported function uniformly by smooth compactly supported functions, hence in $L^2(\mu)$ since μ is finite. Such functions are Schwartz. The same argument applies to real and imaginary parts.

Since $D_0 \subset D_{\mathrm{w}}$ and the latter has closed single-valued graph, D_0 is closable. Explicitly if $f_n \to 0$ and $f_n' \to v$ in H, their graph pairs tend to $(0, v)$ in $\mathcal{G}$, so $v = 0$. The closure of the core graph therefore defines a densely defined closed operator D with

$$D_0 \subset D \subset D_{\mathrm{w}}. \tag{6}$$

This is an inclusion, not a proved identification of the minimal graph closure with every distributional solution of (5). The Schwartz graph is a core for D by its definition. No different maximal weighted Sobolev domain or condition at individual zeros is substituted for it.

The nonnegative core form $\kappa\|D_0 f\|^2$ is closable for every $\kappa \geq 0$. For $\kappa > 0$, a sequence tending to zero whose derivative energies are Cauchy has derivatives Cauchy in H, hence convergent by completeness, and the preceding graph test makes their limit zero. For $\kappa = 0$ the assertion is immediate. This is positive *absolute-density* energy, not an energy obtained by integrating a square against the signed law.

4 Explicit adjoint probes and their totality

The real Hilbert adjoint of D is defined by

$$v \in \operatorname{dom} D^* \iff \exists z \in H \ \forall u \in \operatorname{dom} D : \ \langle v, Du\rangle = \langle z, u\rangle, \qquad D^* v = z.$$

Density of $\operatorname{dom} D$ makes z unique. Its graph is closed by passing these inner-product identities to limits. For every complex Schwartz test φ define explicit H vectors

$$p_\varphi = w\overline{\varphi}, \qquad q_\varphi = -w\overline{\varphi'} - b\overline{\varphi}. \tag{7}$$

Boundedness of the coefficients and finiteness of μ put both in H. Their normalization is

$$\langle p_\varphi, u\rangle = \operatorname{Re} Eu(\varphi), \quad \langle q_\varphi, u\rangle = -\operatorname{Re} Eu(\varphi') - \operatorname{Re} Fu(\varphi).$$

For $(u, v) \in \mathcal{G}$, (5) therefore gives $\langle p_\varphi, v\rangle = \langle q_\varphi, u\rangle$. Applied to $v = Du$, this proves

$$p_\varphi \in \operatorname{dom} D^*, \qquad D^* p_\varphi = q_\varphi. \tag{8}$$

The formal development defines these vectors by real Riesz representation; (7) is the direct coordinate realization verified by the two displayed pairings. No division by a vanishing density is needed.

Their real linear span is dense. If u is orthogonal to every p_φ, then $\operatorname{Re} Eu(\varphi) = 0$ for every φ. Testing $i\varphi$ gives the imaginary part as well, so $Eu = 0$ and injectivity gives $u = 0$. The closed span has zero orthogonal complement and is therefore H. In particular $\operatorname{dom} D^*$ is dense. These probes are total in H; they have not been asserted to be a graph core for D^*.

For completeness the bidual is the original D. A pair (u, v) is in its closed graph precisely when it is orthogonal to every vector of its graph's orthogonal complement. The latter consists of $(-D^* z, z)$ for $z \in \operatorname{dom} D^*$: orthogonality to (f, Df) is exactly the adjoint definition. Orthogonality of (u, v) to all these vectors is $\langle v, z\rangle = \langle u, D^* z\rangle$, the definition of (u, v) in the graph of D^{**}. Thus $D^{**} = D$, without replacing the original domain by $\operatorname{dom} D_{\mathrm{w}}$.

5 The signed form and its full adjoint domain

Multiplication by σ gives a bounded real-linear involution J: $J^2 = I$, $\|Ju\| = \|u\|$, and $\langle Ju, v\rangle = \langle u, Jv\rangle$. These follow pointwise from $\sigma^2 = 1$ and the fact that σ is real. The signed form

$$[u, v] = \langle Ju, v\rangle = \operatorname{Re} \int \overline{u} v K \, dx$$

is continuous and symmetric, and $[u, Ju] = \|u\|^2$. It is nondegenerate even if $[u, u] = 0$ for nonzero u; isotropy is different from degeneracy. The Hilbert space decomposes into the closed functions supported on $\{K > 0\}$ and $\{K < 0\}$, where this form is respectively positive and negative. The zero set has zero μ measure. This is the familiar real indefinite Hilbert-space construction by a fundamental symmetry.

Suppose $v, z \in H$. The identity $[Du, v] = [u, z]$ for every $u \in \operatorname{dom} D$ is equivalent, using symmetry of J, to

$$\langle Du, Jv\rangle = \langle u, Jz\rangle \quad \text{for every } u \in \operatorname{dom} D.$$

By the Hilbert adjoint definition this holds if and only if $Jv \in \operatorname{dom} D^*$ and $D^* Jv = Jz$. Multiplying by J proves the exact domain and formula $D^+ v = J D^* J v$ in Theorem 1.1. Its graph is the inverse image of the closed graph of D^* under the continuous invertible map $(v, z) \mapsto (Jv, Jz)$, so it is closed. Its domain is $J(\operatorname{dom} D^*)$, dense because J is a Hilbert isometry onto H and the adjoint domain is dense.

In particular Jp_φ is in the signed-adjoint domain with $D^+(Jp_\varphi) = Jq_\varphi$; these are actual dense test-generated vectors. The signed adjoint is not obtained by presuming a formal logarithmic derivative acts on every Schwartz input. Behavior at zeros must satisfy its actual transported Hilbert-adjoint domain.

6 Graph projection and the exact energy minimizer

The graph of D is a closed subspace of the product Hilbert space $H \oplus H$ with norm squared $\|u\|^2 + \|v\|^2$. Project $(x, 0)$ orthogonally onto it and write the projection as (Rx, Gx). Thus $Rx \in \operatorname{dom} D$ and $Gx = DRx$. Orthogonal projection and the two coordinate maps are bounded linear, so R, G are bounded real-linear maps. Orthogonality of the residual gives, for every $u \in \operatorname{dom} D$,

$$\langle Rx, u\rangle + \langle Gx, Du\rangle = \langle x, u\rangle. \tag{9}$$

The adjoint definition now yields

$$Gx \in \operatorname{dom} D^*, \qquad D^*Gx = x - Rx.$$

Hence $Rx \in \operatorname{dom} D^*D$ and $(I + D^*D)Rx = x$. Taking $u = Rx$ in (9) gives

$$\|Rx\|^2 + \|Gx\|^2 = \langle x, Rx\rangle \geq 0. \tag{10}$$

Projection is contractive, so $\|Rx\| \leq \|x\|$. Self-adjointness of the bounded graph projection also gives $\langle Rx, y\rangle = \langle x, Ry\rangle$ by testing the two product inputs $(x, 0)$ and $(y, 0)$. Thus R is symmetric and positive in the Hilbert quadratic-form sense.

If $Rx = 0$, single-valuedness of the graph gives $Gx = 0$. Equation (9) then says x is orthogonal to the dense domain of D, so $x = 0$. Thus R is injective. If y is orthogonal to its range, symmetry implies $\langle x, Ry\rangle = 0$ for every x, hence $Ry = 0$ and $y = 0$. The range of R is consequently dense.

The orthogonal residual $(x - Rx, -Gx)$ is perpendicular to $(u - Rx, Du - Gx)$. Pythagoras applied to their difference gives the exact gap (1). The last two terms there are nonnegative, so Rx minimizes $\|x - u\|^2 + \|Du\|^2$. If equality holds for another input, $\|u - Rx\|^2 = 0$ gives uniqueness. Both its domain membership and its derivative value have been established before writing this minimization statement.

7 Regularization preserves the original domain boundary

The construction has a useful convergence consequence which does not extend the native derivative. We give the argument for any closed densely defined real Hilbert operator $T : E \supset \operatorname{dom} T \to F$; in particular it applies to the original closed derivative D above. Put $\varepsilon_n = 2^{-n}$, and project $(x, 0)$ onto the graph of $\varepsilon_n T$. Write its first coordinate as $R_n^T x$, and set $V_n^T x = T R_n^T x$, the *unscaled* native output. The same projection argument gives

$$\langle R_n^T x, u\rangle + \varepsilon_n^2 \langle V_n^T x, Tu\rangle = \langle x, u\rangle \qquad (u \in \operatorname{dom} T), \tag{11}$$

$$\|x - R_n^T x\|^2 + \varepsilon_n^2 \|V_n^T x\|^2 \leq \|x - u\|^2 + \varepsilon_n^2 \|Tu\|^2. \tag{12}$$

Taking $u = R_n^T x$ in the weak equation yields $\|R_n^T x\|^2 + \varepsilon_n^2 \|V_n^T x\|^2 = \langle x, R_n^T x\rangle$. Thus $\|R_n^T x\| \leq \|x\|$, and also $\varepsilon_n \|V_n^T x\| \leq \|x\|$. For any fixed $x \in E$, choose $u \in \operatorname{dom} T$ arbitrarily close to x; then let $n \to \infty$ in (12). This proves

$$R_n^T x \longrightarrow x \quad \text{for every } x \in E, \qquad \varepsilon_n^2 V_n^T x \longrightarrow 0.$$

The second assertion follows from the bound $\varepsilon_n^2 \|V_n^T x\| \leq \varepsilon_n \|x\|$. It does not assert convergence of $V_n^T x$.

`ScaledGraphResolvent`:

`regularize_mem`, `native_weak_identity`, `regularize_tendsto`, `weighted_gradient_tendsto_zero`.

We recall the required adjoint graph fact directly. For a densely defined T, its product-space graph satisfies

$$\operatorname{graph}(T)^{\perp} = \{(-T^*w, w) : w \in \operatorname{dom} T^*\}.$$

Indeed orthogonality says precisely $\langle w, Tu\rangle = \langle -z, u\rangle$ for a pair (z, w) and every $u \in \operatorname{dom} T$, the definition of the adjoint. If $y \perp \operatorname{dom} T^*$, then $(0, y)$ is perpendicular to this orthogonal graph and so belongs to the original closed graph. Single-valuedness forces $y = T0 = 0$. Hence $\operatorname{dom} T^*$ is dense. The same graph identity also proves the following test: (x, v) belongs to the original graph if and only if $\langle x, T^*w\rangle = \langle v, w\rangle$ for every $w \in \operatorname{dom} T^*$. Thus the graph is recovered from all its adjoint tests, without assuming an unproved maximal differential expression.

`HilbertAdjointDomainDensity`:
`closed_adjoint_dense`, `closed_bidual`.

The weak equation (11) implies $V_n^T x \in \operatorname{dom} T^*$ and $\varepsilon_n^2 T^* V_n^T x = x - R_n^T x$. If $x \in \operatorname{dom} T$, then for every $w \in \operatorname{dom} T^*$,

$$\langle V_n^T x, w\rangle + \varepsilon_n^2 \langle T^* V_n^T x, T^* w\rangle = \langle R_n^T x, T^* w\rangle + \langle x - R_n^T x, T^* w\rangle = \langle Tx, w\rangle.$$

This is the variational equation for the same resolvent construction applied to T^* with input Tx. That solution is unique: subtract two weak equations and test against their difference to obtain the sum of a norm square and ε_n^2 times another norm square equal to zero. Therefore

$$TR_n^T x = R_n^{T^*}(Tx), \qquad \|TR_n^T x\| \le \|Tx\|, \qquad TR_n^T x \longrightarrow Tx \quad (x \in \operatorname{dom} T).$$

The adjoint is closed because its defining pairing identities persist under convergence of both graph coordinates. The last limit therefore uses the already proved strong identity approximation for that closed densely defined adjoint. This proves convergence in the *original* graph norm on its original domain.

`ScaledGraphResolventDomain`:
`nativeGradient_dual_regularize`, `nativeGradient_norm_le`,
`nativeGradient_tendsto`, `original_graph_tendsto`.

Proposition 7.1 (Escape outside the native domain)**.** *If $u_j \in \operatorname{dom} T$ converge strongly to $x \notin \operatorname{dom} T$, then $\|Tu_j\| \to \infty$. In particular $\|TR_n^T x\| \to \infty$ for such an x, despite strong convergence of the regularized values.*

Proof. Otherwise some infinite subsequence has $\|Tu_j\| \le C$ for a fixed $C \ge 0$. For $w \in \operatorname{dom} T^*$, pass to the limit along it in $\langle u_j, T^*w\rangle = \langle Tu_j, w\rangle$. Cauchy–Schwarz gives $|\langle x, T^*w\rangle| \le C\|w\|$. This bounded linear functional on the dense subspace $\operatorname{dom} T^* \subset F$ extends continuously to F; the Hilbert Riesz theorem represents it as $\langle v, w\rangle$ for some $v \in F$. The adjoint graph test just proved places (x, v) in the graph of T, a contradiction. Consequently only finitely many indices can have $\|Tu_j\| \le C$ for any finite C, exactly the asserted divergence. □

`ClosedOperatorEscapeEnergy`:
`mem_domain_of_frequently_bounded_graph`,
`output_norm_tendsto_atTop`.

There is a complementary bounded-output conclusion. If $u_j \to x$ and $\sup_j \|Tu_j\| \le C$, the same functional argument first shows $x \in \operatorname{dom} T$. The adjoint pairings converge to those of Tx. For arbitrary $w \in F$, approximate w by $z \in \operatorname{dom} T^*$ and use

$$|\langle Tu_j - Tx, w - z\rangle| \le (C + \|Tx\|)\|w - z\|.$$

Letting $j \to \infty$ and then $z \to w$ proves $Tu_j \rightharpoonup Tx$. Strong convergence of outputs does not follow. For example, on real ℓ^2 take the dense closed diagonal operator $(Tx)_j = jx_j$, with domain $\sum_{j\geq 1} j^2 x_j^2 < \infty$. Finite sequences give density; coordinatewise convergence of a converging graph sequence identifies its closed graph. For its unit vectors, $u_j = e_j/j \to 0$ but $Tu_j = e_j$ has norm one and converges only weakly to zero, since every ℓ^2 coordinate sequence tends to zero. This last example is an elementary human illustration, not a separate mapped formal counterexample.
`ClosedOperatorWeakGraphLimit`:
`adjoint_pairing_tendsto`, `pairing_tendsto`, `weak_tendsto`.

These classical Hilbert facts apply to the minimal signed-density derivative already constructed. They do not identify it with the weak extension, repair a nonclosable atomic derivative, or turn a vanishing weighted output into a native gradient at a point outside its domain.

8 The full Hilbert energy operator

We finish the operator part without invoking an unspecified maximal differential expression. If $u \in \operatorname{dom} D$ and $v = Du \in \operatorname{dom} D^*$, set $z = D^*v$ and $x = u + z$. For every $a \in \operatorname{dom} D$, the adjoint identity gives $\langle x, a\rangle = \langle u, a\rangle + \langle v, Da\rangle$. Comparing with (9) and testing the difference against $a = u - Rx$ yields

$$\|u - Rx\|^2 + \|v - Gx\|^2 = 0.$$

Thus $u = Rx$ and $z = x - Rx$. Conversely every Rx has exactly this factorization by the previous section. Therefore

$$\operatorname{dom} A = \operatorname{ran} R, \qquad A(Rx) = x - Rx, \qquad A = D^*D \tag{13}$$

with the precise product domain in Theorem 1.1. Injectivity of R makes this single-valued and density of its range makes A densely defined. Its quadratic form is $\langle Au, u\rangle = \|Du\|^2 \geq 0$.

Symmetry follows either from D^*D or directly from symmetry of R: $\langle x - Rx, Ry\rangle = \langle Rx, y - Ry\rangle$. To prove full self-adjointness, let $z \in \operatorname{dom} A^*$ with $A^*z = y$. Testing against Rx for arbitrary $x \in H$ gives

$$\langle x - Rx, z\rangle = \langle Rx, y\rangle.$$

Symmetry of R makes this $\langle x, z - Rz - Ry\rangle = 0$ for all x. Hence $z = R(z + y)$, so $z \in \operatorname{ran} R = \operatorname{dom} A$ and $Az = z + y - R(z + y) = y$. Thus $A^* \subseteq A$; symmetry gives the reverse inclusion. An adjoint is closed by its defining pairings, so $A = A^*$ is also closed. Equation (13) proves that $I + A$ is onto and its inverse is exactly the constructed R. This completes every operator assertion of Theorem 1.1.

The weak distributional factorization retained by these actual solutions is

$$E(DRx) = (E(Rx))' - F(Rx), \qquad D^*(DRx) = x - Rx.$$

It does not identify A with an unqualified pointwise Sturm–Liouville expression or identify an energy-semigroup law with the signed density $K\,dx$.

9 An auxiliary Gaussian field with Green covariance

The positive Hilbert resolvent also yields an actual auxiliary Gaussian field, even though the original density may change sign. This construction uses the real Hilbert space $H = L^2(|K|\,dx; \mathbb{C})_{\mathbb{R}}$ from Section 1, with its real inner product. Its index $x \in H$ is a deterministic Hilbert-space label. Neither $K\,dx$ nor a signed square is used as the Gaussian probability measure.

Write $\mathcal{G} = H \oplus H$ with the Hilbert sum norm, and define the bounded real-linear feature map

$$\Phi x = (Rx, Gx) = (Rx, DRx) \in \mathcal{G}.$$

It is exactly the orthogonal projection of $(x, 0)$ onto the original closed graph. Thus

$$\langle \Phi x, \Phi y \rangle_{\mathcal{G}} = \langle x, Ry \rangle, \qquad \|\Phi x\|^2 = \langle x, Rx \rangle \le \|x\|^2. \tag{14}$$

For the first identity, Φx is perpendicular to $(y, 0) - \Phi y$, so $\langle \Phi x, \Phi y \rangle = \langle \Phi x, (y, 0) \rangle = \langle Rx, y \rangle = \langle x, Ry \rangle$. Contractivity of projection proves the bound. The map is injective: $\Phi x = 0$ implies $Rx = 0$, and the dense-domain argument in Section 6 then gives $x = 0$. Consequently $\langle x, Rx \rangle > 0$ for every nonzero label, without asserting a uniform positive lower bound.

The real Hilbert space H is separable. For instance, finite rational step functions on rational intervals, with rational real and imaginary parts, give a countable dense family after the usual truncation and finite-Borel-measure approximation in $L^2(|K|\,dx)$. Hence $\mathcal{G}$ has a finite or countable orthonormal basis (e_j). On a separate product probability space take independent standard real Gaussians (g_j) and set

$$X_x = \sum_j \langle \Phi x, e_j \rangle g_j \quad \text{in } L^2(\mathbb{P}). \tag{15}$$

If the Hilbert space is zero, take the empty sum. Parseval makes every series Cauchy in L^2, supplies an actual measurable random variable up to almost-everywhere equality and gives a bounded real-linear map $x \mapsto X_x$ from H into $L^2(\mathbb{P})$.

Theorem 9.1. *The family $(X_x)_{x \in H}$ is a centered real Gaussian field and*

$$\begin{aligned} \operatorname{Cov}_{\mathbb{P}}(X_x, X_y) &= \langle x, Ry \rangle, \\ \mathbb{E} X_x^2 &= \langle x, Rx \rangle > 0 \qquad (x \neq 0). \end{aligned} \tag{16}$$

For every $u \in \operatorname{dom} A$ and $x \in H$, its exact Green identity is

$$\operatorname{Cov}_{\mathbb{P}}(X_x, X_{u+Au}) = \langle x, u \rangle. \tag{17}$$

All covariance products are integrable. The linearity statements concern actual L^2 classes; they do not assert simultaneous pointwise linearity for every label on one common null-set complement.

Proof. Finite partial sums have mean zero and covariance the corresponding finite coefficient pairing. Their L^2 limits retain zero mean, and Cauchy–Schwarz passes the products in L^1, proving the covariance formula by Parseval and (14). For any finite labels $x_1, \dots, x_k$ and real coefficients a_i, $\sum_i a_i X_{x_i} = X_{\sum_i a_i x_i}$ in L^2. Its partial sums have characteristic function $\exp(-t^2 \sum_{j \le n} \langle \Phi \sum_i a_i x_i, e_j \rangle^2 / 2)$. The inequality $|e^{itv} - e^{itw}| \le |t||v - w|$ passes these expectations to the limit, identifying a centered normal law of variance $\|\Phi \sum_i a_i x_i\|^2$. This proves joint Gaussianity by all real linear combinations, including degenerate combinations. Injectivity proves the strict variance statement. Finally the actual resolvent identity $R(u + Au) = u$, with u in the full product domain already proved, turns (16) into (17). □

`EnergyResolventGaussianField`:
`feature_inner`, `field_gaussian`, `field_covariance`, `field_law`, `variance_positive`.
`SignedSchwartzGaussianField`:
`field_gaussian`, `field_covariance`, `variance_positive`, `green_covariance`.

Original auxiliary Gaussian calculus and its limits

The same construction gives a precise first-chaos derivative and adjoint. Identify the coefficients of Φx with $a(x) = (\langle \Phi x, e_j \rangle)_j \in \ell^2$. On the auxiliary Gaussian product, let $\mathcal{D}$ be the original closed Gaussian derivative and $\delta_{\mathrm{aux}} = \mathcal{D}^*$ its Hilbert adjoint. Then

$$X_x \in \operatorname{dom} \mathcal{D}, \quad \mathcal{D}X_x = a(x), \qquad a(x) \in \operatorname{dom} \delta_{\mathrm{aux}}, \quad \delta_{\mathrm{aux}} a(x) = X_x. \tag{18}$$

The vector $a(x)$ is a constant random field, so $\|\mathcal{D}X_x\|^2_{L^2(\mathbb{P};\ell^2)} = \langle x, Rx \rangle$. For a proof, finite-coordinate linear sums have gradient their constant coefficient vectors. Parseval gives strong convergence of both values and gradients, and closedness supplies the first graph. For the second graph, finite-dimensional Gaussian integration by parts gives $\mathbb{E}[f X_x] = \mathbb{E}\langle \mathcal{D}f, a(x) \rangle$ for smooth cylindrical f by passing the finite coefficient sums in L^2. Approximation in the original Gaussian derivative graph extends it to every $f \in \operatorname{dom} \mathcal{D}$, establishing the full adjoint identity. The canonical inverse-number recovery is also $a(x)$: on first chaos the number eigenvalue is one, so no further inverse weighting occurs. This last identification is the source's first-chaos recovery, not a statement of uniqueness among all possible divergence preimages.

`GaussianFirstChaosRecovery`:
`firstChaos_derivative_graph`, `firstChaos_divergence_graph`, `recovery_firstChaos`.
`SignedSchwartzGaussianField`:
`source_calculus`.

These are auxiliary Gaussian source operators. They are not the original spatial derivative D on $L^2(|K|\,dx)$; the construction supplies no intertwining that identifies those two derivatives. The exact resolvent covariance is the connection that has been proved. It applies, in particular, to each actual higher even heat Schwartz marginal and to the normalized simple-zero example in the next section, regardless of failure of a square-integrable logarithmic score.

Nor has an H-valued Gaussian random variable with covariance R been constructed. A Hilbert-indexed field needs no trace-class hypothesis. For the general closed-derivative construction this distinction is necessary: take $D = 0$ on an infinite-dimensional Hilbert space. Then $R = I$ and the indexed values on an orthonormal basis are independent standard Gaussians. If they were coordinates of a Hilbert-valued sample, their squared sum would be finite. But the independent events $\{g_j^2 \geq 1\}$ have a common positive probability, so the independent Borel–Cantelli lemma gives infinitely many of them almost surely, and that sum diverges. This is a human general counterexample, not a claim that the specific signed-density resolvent equals I.

Nualart [4], Section 3.4, gives the classical isonormal construction. The graph-projection feature above realizes its exact resolvent covariance directly; no new general Gaussian representation principle is claimed. In particular auxiliary positivity neither removes the sign of K nor builds a positive path law for a signed pseudoprocess.

10 An integrable score, simple zeros and a normalized example

Define $s_K = -K'/K$ where $K \neq 0$ and set it to zero where $K = 0$, the division convention of the source. The cancellation $K s_K = -K'$ holds Lebesgue almost everywhere, including the zero set up to a null subset. To see the last qualification, every zero where $K' \neq 0$ is isolated among zeros, by the definition of a nonzero derivative. A discrete subset of $\mathbb{R}$ is countable: choose for each isolated point a rational interval containing no other point, and use the countable rational basis. Thus $K' = 0$ at almost every zero of K. Consequently

$$\int |s_K|\,d\mu = \int |K'|\,dx < \infty. \tag{19}$$

For every Schwartz test f, ordinary integration by parts gives the actual signed Stein identity

$$\int f'K\,dx = \int f s_K K\,dx.$$

The right side is absolutely integrable since f is bounded and (19) holds. Existence of this signed integral does not imply the score belongs to H.

Suppose $K(0) = 0$ and $K'(0) \neq 0$. Continuity of K' and the derivative definition give $c, C, \delta > 0$ such that, for $0 < x < \delta$,

$$|K'(x)| \geq c, \qquad 0 < |K(x)| \leq Cx.$$

Indeed the first inequality follows by staying within half of $|K'(0)|$; the quotient $K(x)/x \to K'(0)$ gives both nonvanishing and the upper bound for sufficiently small x. Then

$$\int |s_K|^2\,d\mu \geq \int_0^\delta \frac{|K'(x)|^2}{|K(x)|}\,dx \geq \frac{c^2}{C}\int_0^\delta \frac{dx}{x} = \infty.$$

Thus a transverse zero obstructs L^2 score integrability, while all of Theorem 1.1 still holds.

For a concrete normalized signed density set

$$K_0(x) = \frac{(x + x^2)e^{-x^2}}{M}, \qquad M = \int_{\mathbb{R}} (x + x^2)e^{-x^2}\,dx > 0.$$

The odd term integrates to zero by $x \mapsto -x$. The remaining $x^2e^{-x^2}$ is nonnegative and strictly positive on $(1, 2)$, so its integral is positive and finite. Hence K_0 is Schwartz, has integral one and is strictly negative at $x = -1/2$. Also $K_0(0) = 0$ and $K_0'(0) = 1/M \neq 0$. It follows that $s_{K_0} \in L^1(|K_0|\,dx) \setminus L^2(|K_0|\,dx)$. Its original derivative closure, dense signed adjoint and positive Hilbert energy resolvent/operator nevertheless exist. This counterexample does not say the score of every sign-changing higher-heat kernel has a simple zero; that extra zero hypothesis has not been proved or assumed here.

11 Polynomial Stein weights and higher even heat marginals

There is a different first-order obstruction that needs no simple-zero hypothesis. Suppose a real polynomial P satisfies

$$\int f'K\,dx = \int fPK\,dx \quad \text{for every real Schwartz } f.$$

Integration by parts says $\int f(K' + PK) = 0$ for all such f. The residual $K' + PK$ is itself Schwartz; choosing f to be that residual gives its squared integral zero. Continuity then gives $K' + PK = 0$ at every point. If $Q(x) = \int_0^x P(y)\,dy$, differentiation yields

$$(e^{Q(x)}K(x))' = e^{Q(x)}(K'(x) + P(x)K(x)) = 0.$$

The expression is constant on $\mathbb{R}$ by the mean value theorem. Since the exponential is strictly positive, a nonzero such K has one fixed sign. Thus a sign-changing real Schwartz density cannot have any polynomial first-order Stein weight of this form. This does not preclude the measurable L^1 score above, higher derivative identities or a calculus on a specified test dual.

For example let $p = 2m > 2$ and $t > 0$, and let k_t be the real even Schwartz kernel with angular Fourier transform $\widehat{k_t}(\xi) = e^{-t\xi^p}$. Its mass is one and its second moment is zero, by differentiating

that transform twice at zero. If $k_t \geq 0$ almost everywhere, the nonnegative function $x^2 k_t(x)$ would have zero integral, forcing $k_t = 0$ away from zero almost everywhere, contradicting its unit Lebesgue mass. Thus it has a negative value; positive mass also gives a positive value. It is sign-changing, so the polynomial first-order weight is impossible. On the other hand its absolute-density space has every closed operator constructed above. The formal library uses Fourier phases $e^{2\pi i\xi x}$; rescaling angular frequency changes neither zero second moment nor sign and operator conclusions.

This is a statement about each fixed signed marginal. It gives neither a finite-variation signed path law nor predictable representation, and its positive energy is not a signed-square isometry. The separate completed multichannel test-dual calculus has different domains and must retain them.

12 Full support does not replace the density hypothesis

The preceding theorem uses an absolutely continuous reference $|K|\,dx$. A different, positive probability measure can have full support and a well-defined smooth derivative but no closable extension of that derivative. Let γ be the standard real Gaussian law and put

$$\widetilde{\mu} = \tfrac{1}{2}\gamma + \tfrac{1}{2}\delta_0.$$

It is a finite Radon probability with full support, since every nonempty open interval has positive Gaussian mass. Its atom at zero has mass $1/2$. Consider the ordinary real derivative on $C_c^\infty(\mathbb{R})$ in the real space $L^2(\widetilde{\mu})$. It is well-defined: if two continuous representatives agree $\widetilde{\mu}$-almost everywhere, they agree everywhere. Otherwise continuity gives a nonempty open interval where their difference stays nonzero, contradicting full support. Their ordinary derivatives therefore agree. The domain is dense by finite Radon regularity and mollification, just as in Section 3.

Choose a smooth compact cutoff χ equal to one for $|x| \leq 1$ and zero for $|x| \geq 2$. Such a cutoff is explicit: take $\eta(s) = e^{-1/s}$ for $s > 0$, zero for $s \leq 0$, and set

$$\chi(x) = \frac{\eta(4 - x^2)}{\eta(4 - x^2) + \eta(x^2 - 1)}.$$

The denominator is everywhere positive. Smoothness at the seams follows because every derivative of $e^{-1/s}$ is a finite sum of polynomial powers of $1/s$ times $e^{-1/s}$ and tends to zero as $s \downarrow 0$. Thus $\chi \in C_c^\infty$ with the specified values. Write $\psi(x) = x\chi(x)$ and

$$f_n(x) = \frac{\psi((n+1)x)}{n+1}, \qquad f_n'(x) = \psi'((n+1)x).$$

Then $f_n(0) = 0$, $f_n'(0) = 1$ and $\sup|f_n| \leq \sup|\psi|/(n+1) \to 0$. For each $x \neq 0$, compact support makes $f_n'(x) = 0$ eventually, whereas at zero it remains one. The derivatives are uniformly bounded by $\sup|\psi'|$. Dominated convergence for the finite measure therefore gives

$$f_n \longrightarrow 0, \qquad f_n' \longrightarrow 1_{\{0\}} \quad \text{in } L^2(\widetilde{\mu}), \qquad \|1_{\{0\}}\|_2^2 = \tfrac{1}{2}. \tag{20}$$

The graph has a nonzero vertical limit. Hence no closable linear operator agreeing with all these smooth derivative data can exist: its graph closure would contain both $(0,0)$ and $(0, 1_{\{0\}})$ and would not be single-valued.

The positive derivative form also fails closability. For $\kappa > 0$, (20) implies

$$\kappa\|f_n'\|_2^2 \longrightarrow \kappa/2 > 0, \qquad \kappa\|f_n' - f_m'\|_2^2 \longrightarrow 0 \quad (n, m \to \infty),$$

although $f_n \to 0$ in the underlying space. This violates the defining sequential form-closability test. At $\kappa = 0$ the zero form is trivially closable. Full support and uniqueness of continuous representatives do not repair this graph obstruction.

For the pure Dirac law the defect occurs earlier: ψ and the zero function have the same $L^2(\delta_0)$ class since $\psi(0) = 0$, but their derivatives have different classes since $\psi'(0) = 1$. Thus ordinary smooth differentiation does not even descend to those equivalence classes. The Gaussian mixture restores representative independence while preserving nonclosability. These are separate positive-measure examples, not failures of Theorem 1.1 for a Schwartz absolute density.

13 Finite restrictions and a sufficient compatibility criterion

The mixed-measure example also shows why checking finite restrictions alone cannot prove closability. Let $\mathcal{D}$ be the real smooth derivative domain in $X = L^2(\widetilde{\mu}; \mathbb{R})$ just used. For every finite-dimensional subspace $W \subset \mathcal{D}$, the linear map $D|_W : W \to X$ is continuous in the norm induced from X, by finite-dimensional norm equivalence. Thus $u_n \in W$, $u_n \to 0$ implies $Du_n \to 0$. Its derivative form is also closable on W: indeed its values tend to zero without even needing the energy-Cauchy assumption. Here closability refers to the sequential test on W; W need not be dense in X.

These windows are directed under sums and cover $\mathcal{D}$, since every u belongs to its one-dimensional span. Nevertheless (20) proves that their union derivative and every positively weighted derivative form fail closability. No fixed finite window contains infinitely many of those f_n: otherwise its subsequence would have derivatives tending both to zero by continuity and to $1_{\{0\}}$ by (20). This last observation is a classical human deduction, not an additional formal theorem. The issue is compatibility across windows, not failure of any one finite-dimensional check.

Here is a precise positive criterion from the same frozen library. Let V be a real vector space, X a normed space and Y a Banach space, with linear maps $I : V \to X$ and $D : V \to Y$. Sequential closability means

$$Iu_n \to 0, \quad Du_n \to y \quad \Longrightarrow \quad y = 0.$$

The map I need not be injective. Suppose a family of linear maps $P_i : V \to V$ and bounded linear maps $Q_i : X \to X$ satisfies

$$IP_i = Q_i I, \qquad \|DP_i v\| \le \|Dv\| \quad (v \in V), \tag{21}$$

$$\text{for every } v \in V \text{ some } i \text{ has } P_i v = v, \tag{22}$$

and each pair (IP_i, DP_i) is sequentially closable. Then (I, D) is sequentially closable. No nesting, countability, common bound on Q_i, or idempotence of P_i is required by this statement. The exact energy contraction in (21) is an assumption, not a consequence of finite-dimensionality.

Proof. Suppose $Iu_n \to 0$ and $Du_n \to y$. For each fixed i, boundedness of Q_i implies $IP_i u_n \to 0$, while

$$\|DP_i u_n - DP_i u_m\| \le \|Du_n - Du_m\|$$

makes the projected derivatives Cauchy. Completeness gives a limit, and local closability forces that limit to be zero. For fixed $v \in V$, choose i with $P_i v = v$. Contraction applied to $u_n - v$ gives

$$\|DP_i u_n - Dv\| \le \|Du_n - Dv\|.$$

Take $n \to \infty$ to obtain $\|Dv\| \le \|y - Dv\|$. This holds for every v, so substitute $v = u_m$ and then let $m \to \infty$. The left side tends to $\|y\|$ and the right side to zero; hence $y = 0$. □

For $\kappa > 0$, the form $\mathcal{E}_\kappa(v) = \kappa\|Dv\|^2$ is closable relative to I exactly when (I, D) is sequentially closable, provided Y is complete. In one direction, an energy-Cauchy sequence has Du_n Cauchy; its limit is zero by operator closability, so $\mathcal{E}_\kappa(u_n) \to 0$. In the other, if $Du_n \to y$ and $Iu_n \to 0$, then $\mathcal{E}_\kappa(u_n - u_m) \to 0$; form closability gives $\kappa\|Du_n\|^2 \to 0$, forcing $y = 0$. At $\kappa = 0$ the zero form is always closable and gives no operator information. Thus (21)–(22) also yield closability of every nonnegative weighted derivative form. The finite-window counterexample rules out a compatible family with all these hypotheses on its original nonclosable domain; a covering family of finite subspaces alone supplies none of the needed energy and input compatibility.

14 Classical antecedents and exact source boundary

Weighted derivative closability is classical. De Cicco–Serra Cassano [1], Section 2.2/condition (12), recall Hamza's one-dimensional criterion. Continuous w is locally bounded below where positive, so $1/w$ is locally integrable there. Their bounded-interval relaxation results do not identify our global minimal and weak graphs. Derkach–Trunk [3], Section 2.1, give the classical $T^+ = JT^*J$ mechanism. Their coupling and spectral theorems are not used. Graph projection and D^*D are classical Hilbert theory; no priority is claimed.

For singular reference measures, Alberti–Bate–Marchese [2], Corollary 1.6, gives a much broader necessity of absolute continuity for classical differential-operator closability in the range $1 \le p < \infty$, $1 \le q \le \infty$. Their domain is C_c^1 and their Lipschitz convergence topology is separately defined. Our smooth shrinking jets provide a direct witness for the particular mixture; we do not derive a new general classification. Finite-dimensional continuity and the compatible-window norm argument above are also presented without a priority claim.

The exact theorem map records frozen declarations at

`bab688bb6140cd6316c80680095a914ea0d346db`.

The map also records the actual smooth shrinking jets and mixed Gaussian/Dirac nonclosability; the explicit cutoff above realizes the same unit derivative and radius-one/two support properties. Riesz-probe coordinates and regrouped human proofs are labelled reconstructions. The original full-build/axiom evidence is hash-checked; no research source or original domain is changed. This formalization companion awaits independent referee review.

References

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[2] G. Alberti, D. Bate and A. Marchese, *On the closability of differential operators*, primary final preprint, May 2025. arXiv:2311.08058v2. Introduction/Corollary 1.6 comparison only.

[3] V. Derkach and C. Trunk, *Coupling of definitizable operators in Kreĭn spaces*, TU Ilmenau preprint M 17/03 (2017). https://d-nb.info/1125462493/34. Section 2.1 comparison only.

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